

How to read this: values are the labeled or published average serum estradiol (pg/mL) at steady state. Absorption varies by patient, skin site, and body composition — the only way to know a given patient's number is a drawn level. There is no official target: The Menopause Society is explicit that hormone therapy is dosed to symptom relief, not to a lab number.

Orientation

Where the levels land

- ≈ 10-25

Ultra-low — bone protection, mild symptom relief for some. Below most premenopausal levels.
- ≈ 26-45

Low — the most common starting range; roughly early-follicular.
- ≈ 46-70

Mid — where many patients land once titrated to full symptom control.
- ≈ 71-100

Higher — the top of standard FDA-approved transdermal dosing.
- > 100

Above standard — at approved doses only oral 2 mg and injections reach here.

For reference, untreated postmenopausal E2 runs about 2–15 pg/mL; premenopausal early follicular 25–75, luteal 50–150, ovulatory peak 150–400.

Transdermal

Patch

Bypasses first-pass liver metabolism — no added clot risk. Prescribe twice-weekly patches; once-weekly reservoirs run low by day 5–6.

Dose	Brands (labeled level)	pg/mL
0.014 mg/day	Menostar (weekly) — bone protection only	14
0.025 mg/day	Climara ~20* · Vivelle-Dot / Dotti ~20* · Alora ~25*	19–25
0.0375 mg/day	Climara ~30* · Vivelle-Dot / Dotti 34	30–34
0.05 mg/day	Climara 40 · Vivelle-Dot / Dotti 57 · Alora 64	40–64
0.06 mg/day	Climara only ~48*	48
0.075 mg/day	Climara ~60* · Vivelle-Dot / Dotti 72 · Alora 86	60–86
0.1 mg/day	Climara 80 · Vivelle-Dot / Dotti 89 · Alora 98	80–98

*Interpolated from the dose-proportionality the label demonstrates. Brands are not reliably interchangeable at the same labeled dose — the spread is real. If one patch isn't working, try another brand before switching routes.

Oral

Oral estradiol

First-pass through the liver raises clotting factors and SHBG, and converts much of the dose to estrone (~5:1 E1:E2 vs. ~1:1 transdermal). Generally the fallback, not the default.

Dose	Note	pg/mL
Estradiol 0.5 mg	Estimated	27
Bijuva 0.5 mg / 100 mg	E2 + micronized progesterone capsule · estimated	17
Bijuva 1 mg / 100 mg		34
Estradiol 1 mg	Reported range 40–66 — widest variance on this page	50
Estradiol 2 mg	Reported range 100–115	110

Transdermal

Gel & spray

Once-daily dosing rises and falls over 24 h (≈1.6–1.7× peak-to-trough) — draw levels at a consistent time after application. Keep the site consistent and let it dry before dressing.

Product / dose	Note	pg/mL
Divigel (gel packets)		
0.25 g (0.25 mg)		10
0.5 g (0.5 mg)		21
0.75 g (0.75 mg)	Extrapolated	26
1.0 g (1.0 mg)		31
1.25 g (1.25 mg)	Extrapolated	38

EstroGel (pump, 0.75 mg per pump)

1 pump		28
2 pumps	Estimated — label publishes 1 pump only	77

Elestrin (pump)

1 pump		15
2 pumps (1.7 g)		39

Evamist (spray, inner forearm)

1 spray		20
2 sprays		31
3 sprays	Plateaus — same level as 2 sprays	31

Alcohol-based gels raise E2 more than a cream at the same labeled dose. Switching between formulations is a real dose change, not a lateral move. If 2 Evamist sprays isn't enough, that points to a different product, not a third spray.

Vaginal

Vaginal estrogen

Low doses work locally and barely move serum E2 — no progestogen needed. Femring is systemic therapy in the same costume.

Product	Dose	pg/mL
Vagifem / Yuvaferm	10 mcg insert	5.5–10
Imvexxy	4 mcg (≈ placebo 4.3)	3.6
Imvexxy	10 mcg	4.6
Estring	7.5 mcg/day ring	7–8.1
Estradiol cream 0.01%	Dose- and technique-dependent	variable
Femring	0.05 mg/day — systemic	41
Femring	0.1 mg/day — systemic	76

Femring needs endometrial protection if she has a uterus. Persistent GSM on systemic therapy is a reason to add a local product, not to raise the systemic dose.

Not on this page: injectable estradiol (cypionate / valerate). Injections have a real peak-and-crash curve and no published steady-state PK tables — modeled peak, trough, and average values for weekly, 2-week, and 4-week intervals are in the app.